

Framing COVID-19 Response and Vaccines Across Ideologies: Evidence from News Headlines

Mia Sager
Emory University
mia.sager@emory.edu

Molly Murphy
Emory University
molly.murphy@emory.edu

Abstract

Political polarization in the United States has increasingly shaped public discourse on health and vaccination, a trend that became especially pronounced during the COVID-19 pandemic. Media outlets played a central role in amplifying these divides. This project examines differences in COVID-19 vaccine coverage across conservative and liberal news sources, focusing on variation in tone, framing, and rhetorical emphasis. Using N-gram analyses, we found clear ideological patterns: liberal outlets emphasized scientific developments, while conservative outlets concentrated more on political figures and conflict. We further observed that polarization in coverage intensified throughout the pandemic, as shown by a yearly logistic regression classifier trained on TF-IDF features, whose accuracy steadily increased from 2020 to 2022. Finally, by comparing vaccine-related language pre-pandemic to post-pandemic, we identified notable shifts in dominant topics and found that overall semantic polarization surprisingly decreased after the pandemic, possibly reflecting the emergence of a more universal, cross-ideological vocabulary surrounding vaccines.

1 Introduction

In recent years, growing political divisions in the United States have made public health and vaccines a deeply polarizing topic. During the COVID-19 pandemic, this divide was especially clear in how people discussed and viewed vaccines and government responses, which is largely driven by the media they consume (Kerr et al., 2021). This paper aims to compare how conservative and liberal media sources covered COVID-19 vaccines and response, focusing on differences in tone, framing, and overall rhetoric, as well as how media coverage on vaccines in general has changed since the pandemic. Our analysis addresses the following research questions:

- 1.) How do conservative and liberal news outlets frame COVID-19 vaccination and response differently?
- 2.) How did these framing patterns evolve over time, from the onset of the pandemic in early 2020 through the vaccine rollout and booster phases in 2022?
- 3.) Have attitudes towards vaccinations (not just COVID-19) in the media significantly shifted since the pandemic?

2 Background

The COVID-19 pandemic quickly became one of the most politically polarized public health events in the United States and around the world. Research in political communication has shown that early conservative and liberal framing of the pandemic and variation in media coverage influenced how audiences understood the severity of the virus and the legitimacy of public health guidelines, shaping threat perception, vaccine hesitancy, and trust in institutions (Motta et al., 2020; Fridman et al., 2021). To quantify such ideological differences, computational linguistics increasingly rely on semantic models. Word2Vec represents words as dense vectors based on contextual co-occurrence, allowing researchers to measure semantic relationships using cosine similarity and compare ideological usage through nearest-neighbor analysis (Mikolov et al., 2013; Gonen et al., 2020). Word embedding has also been used to track semantic change over time, important for analyzing how COVID-related concepts evolved throughout the pandemic, as well as how vaccine-related concepts changed after the pandemic (Hamilton et al., 2016). For supervised learning, logistic regression classifiers, trained on TF-IDF features, are a common approach for identifying partisan language patterns

and distinguishing between ideological sources (Salton and Buckley, 1988; Pedregosa et al., 2011). Log-odds-ratios are used to identify terms that disproportionately occur across groups, and are especially common for ideological language comparisons (Monroe et al., 2017).

Although prior studies examine COVID media framing broadly, few directly compare the semantic structure of conservative and liberal headlines across multiple years, and none combine embedding-based semantic analyses with supervised classification (Sacerdote et al., 2020). Our project addresses this gap by training separate ideological embedding spaces for concepts like “vaccine,” “freedom,” and “mask,” analyzing semantic divergence using cosine similarity and nearest neighbors, and validating results with TF-IDF logistic regression classifiers (Gonen et al., 2020; Salton & Buckley, 1988; Pedregosa et al., 2011). This multi-method framework provides a comprehensive view of ideological polarization in COVID-related headlines from 2020–2022, as well as general vaccine-related headlines before and after the pandemic.

3 Data Overview

We collected data from two publicly available Kaggle datasets. The first contains 4.5 million headlines from the ten largest online news publications from 2007 to 2022. From this, we extracted headlines from three liberal outlets (CNN, The New York Times, The Guardian) and three conservative outlets (Fox News, Daily Mail, New York Post) from 2020–2022, using the AllSides Media Bias Chart for ideological classification (“Media Bias”, 2025) (Appendix A). To examine broader shifts in vaccine attitudes, we gathered 2019 headlines for the same outlets from this dataset and paired them with 2024 headlines from a second Kaggle dataset that scraped Google News articles beginning in late 2023 (Krishnayah, 2023; Crxxom, 2025) (Appendix B). For the COVID-19 analysis, we filtered headlines using keywords such as “covid,” “pandemic,” “lockdown,” “quarantine,” “vaccine,” and “booster” (Appendix C). For the general vaccine-attitudes analysis, we used broader terms like “vaccine,” “vaccination,” “immunization,” and “inoculation.” After filtering, the New York Post contained no vaccine-related headlines in 2024, so it was excluded from the 2019-2024 comparison.

4 Methodology

To address Research Question 1, we trained a logistic regression classifier on TF-IDF features in order to identify the most distinguishable/predictive ngrams (unigrams and bigrams) for liberal and conservative news outlets’ headlines covering the pandemic and vaccines from 2020 to 2022 (Salton and Buckley, 1988; Pedregosa et al., 2011). We also trained separate Word2Vec models on liberal and conservative news outlets’ headlines from 2020 to 2022 (Mikolov et al., 2013). We used these to identify the nearest neighbors of many terms related to the pandemic and vaccines in order to quantify the differences in the framing used by conservative and liberal news outlets when covering these topics (Gonen et al., 2020).

To address Research Question 2, we trained separate Word2Vec models for each year (2020, 2021, and 2022) by ideology (Mikolov et al., 2013). Using these, we explored how the cosine similarities between key vaccine- and covid-related terms changed over time for liberal vs. conservative news outlets, giving insight into how, when, and why framing and rhetoric regarding the pandemic changed (Gonen et al., 2020). Additionally, we trained yearly (2020, 2021, 2022) logistic regression classifiers on TF-IDF features to examine the accuracy of predicting liberal vs. conservative outlets based on covid- and vaccine-related headlines from 2020 to 2022 (Salton and Buckley, 1988; Pedregosa et al., 2011).

To address Research Question 3, we used TF-IDF to identify the highest weighted (most distinctive) terms from vaccine-related headlines pre- and post-pandemic (Salton and Buckley, 1988). Using these TF-IDF weighted terms, we calculated log-odds ratios to identify the terms each ideology disproportionately used when covering vaccines in 2019 vs. 2024 (Monroe et. al). Finally, we trained separate Word2Vec models for liberal and conservative vaccine-related headlines in 2019 and 2024, allowing us to measure the semantic distance between ideologies when covering vaccines pre- and post-pandemic (Mikolov et al., 2013). Semantic distance, calculated as $1 - \text{cosine similarity}$, served as a quantification of the level of polarization between ideologies pre- and post- pandemic, with low semantic distance indicating low polarization and vice versa (Gonen et. al, 2020).

5 Results and Discussion

Using the TF-IDF and logistic regression model, we identified the ngrams that were most predictive of a COVID-related headline coming from a liberal or conservative news outlet during the pandemic, determined by the magnitude of the models' coefficients corresponding to each term (Stalton and Buckley, 1988; Pedregosa et al., 2011). As shown in Figure 1, liberal news outlets tended to focus on scientific updates regarding the situation (e.g., "updates", "latest pandemic", "coronavirus live"). In contrast, conservative news outlets tended to focus on political figures and authority (e.g., "cuomo", "biden", "blasio", "cdc", "gov").

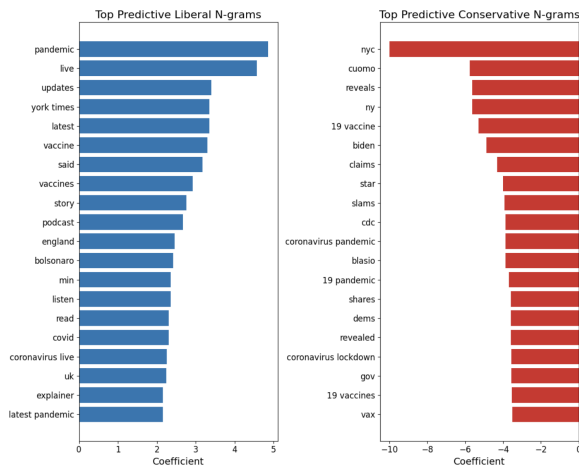


Figure 1: Top predictive n-grams distinguishing liberal and conservative COVID-19 headlines based on TF-IDF features and logistic regression.

By using the separately-trained Word2Vec models by ideology for 2020-2022 to identify the nearest neighbors of vaccine and pandemic related words, calculated by cosine similarity, we were able to establish differences in liberal vs. conservative coverage of these topics (Mikolov et. al, 2013; Gonen et. al, 2020). Interesting differences were found when using this method for the term "mandate," as shown in Figure 2. Liberal outlets associated mandates with administrative and public-health procedures (e.g., "extend," "order," "authority"), while conservative outlets associated it with rights-oriented or oppositional terms (e.g., "unconstitutional," "mandatory," "rules"). This suggests that liberal and conservative sources covered opposing views on government intervention during the pandemic. Applying this method to the term "lockdown" also exposed these opposing views (Appendix D).

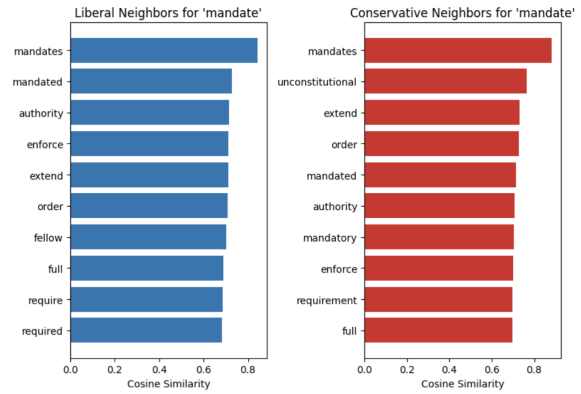


Figure 2: Nearest neighbors of "mandate" in liberal and conservative Word2Vec models, showing contrasting policy and rights framings.

5.1 RQ2: Changes in Framing Over Time (2020-2022)

The separately-trained Word2Vec models by ideology for each year (2020, 2021, 2022) allowed us to explore how and when outlets' coverage on the pandemic changed by comparing cosine similarities for interesting pairs of terms in each year of the pandemic (Mikolov et al., 2013; Gonen et al., 2020). One of the most notable insights gained from this involves freedom-related language regarding vaccines and masks, which is displayed in Figure 3. For "mask" and "freedom", liberal similarity stays stable, while conservative similarity declines sharply after 2021, indicating a shift toward oppositional and polarizing framing/rhetoric. For "vaccine" and "freedom", liberal similarity increases from 2020 to 2021, and then decreases again, while conservative similarity steadily decreases over time. This could be explained by the tendency for liberal outlets to frame vaccination as a way to regain freedom in 2021, while conservative outlets framed freedom more in opposition to vaccination requirements as the pandemic progressed (Bolsen and Palm, 2021). As a result, the semantic meanings of "vaccine" and "freedom" grew farther apart in conservative outlets. Comparing the cosine similarities for the terms "covid" and "risk" and "mask" and "risk" also resulted in interesting insights on the differences in framing between ideologies over time (Appendix E).

Finally, we observed that the accuracies of the yearly logistic regression classifiers with TF-IDF features in predicting whether a covid-related headline was published by a liberal or conservative outlet increased throughout the pandemic (Stalton and

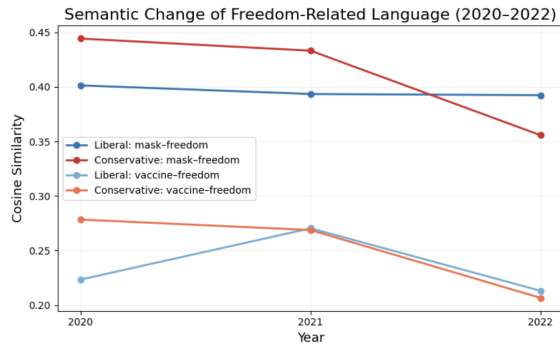


Figure 3: Semantic change (measured by changes in cosine similarities) of freedom-related language from 2020–2022, comparing liberal and conservative similarities for “mask” and “freedom” and “vaccine” and “freedom.”

Buckley, 1988; Pedregosa et al., 2011). The classifiers’ accuracies increase from 0.78 to 0.81 to 0.82 across 2020 to 2022, suggesting that ideological differences in language became more pronounced as the pandemic progressed (Appendix F). This supports the broader pattern of increasing linguistic and semantic polarization seen in the above analyses.

5.2 RQ3: Vaccine Coverage Before vs. After COVID-19

By applying TF-IDF to vaccine-related headlines in 2019 and 2024, we were able to identify the most distinctive terms used when discussing vaccines pre- and post- pandemic (Stalton and Buckley, 1988). We discovered that headlines in 2019 (pre-pandemic) tended to focus on measles outbreaks and the anti-vaccination movement, reflecting concerns about misinformation. 2024 headlines (post-pandemic) instead focus on COVID, flu, polio, and mpox, suggesting that the pandemic resulted in a shift in vaccine coverage away from misinformation and vaccine refusal to ongoing infectious disease threats (Appendix G).

Using these TF-IDF-weighted terms, we calculated log-odds ratios to identify terms used disproportionately by each ideology when covering vaccines in 2019 vs. 2024 (Monroe et al., 2017). Pre-pandemic (2019), liberal outlets concentrated on health measures and community-level impacts (e.g., “treatment”, “risk”, “county”, “groups”), while conservative outlets concentrated on emotional narratives (e.g., “mom”, “son”, “kids”). Post-pandemic (2024), liberal outlets emphasized global outbreaks and scientific-reporting (e.g., “mpox”,

“gaza”, “un”), whereas conservative outlets emphasized skepticism on vaccines (e.g., “died”, “warn”, “illness”) (Appendix H).

Although conservative and liberal outlets certainly focused on different vaccine-related concepts both pre- and post- pandemic, the semantic distances (1 - cosine similarity) between their headlines surprisingly suggest that polarization actually decreased (Gonen et. al, 2020). Using the separately-trained Word2Vec models by ideology for 2019 and 2024, the semantic distance between liberal and conservative vaccine-related headlines decreased from 0.059 in 2019 to 0.048 in 2024, suggesting decreased polarization after the pandemic (Appendix I). We hypothesize that this could be explained by the creation of a more universal vocabulary surrounding vaccines as a result of the pandemic, with terms like “booster”, “variant”, and “dose” becoming more common. This shared vocabulary may have resulted in a decreased semantic distance between liberal and conservative outlets in 2024, even if they still differ in tone and framing.

Limitations

The analyses presented in this paper are limited due to the use of headlines rather than full articles for comparison, which could have resulted in a lack of context considered. Furthermore, focusing on a few major news outlets from each ideology may have excluded diverse perspectives, and the keyword filtering to retrieve the headlines may have excluded some relevant headlines. Thus, the findings presented in this paper should be interpreted as indications of patterns in media reporting, not as definitive measures of differences and bias.

Conclusion

Through our analyses, we found clear ideological patterns: liberal outlets emphasized scientific developments, while conservative outlets concentrated more on opposing political figures. We further observed that polarization in coverage intensified and became headlines more distinguishable by ideology throughout the pandemic. Finally, by comparing vaccine-related language from 2019 (pre-pandemic) to 2024 (post-pandemic), we identified notable shifts in dominant topics and found that overall semantic polarization surprisingly decreased after the pandemic, possibly reflecting the emergence of a more universal, cross-ideological vocabulary surrounding vaccines.

References

AllSides. (2025). Media bias chart. <https://www.allsides.com/media-bias/media-bias-chart>

Bolsen, T., & Palm, R. (2021). Politicization and COVID-19 vaccine resistance in the U.S. *Progress in Molecular Biology and Translational Science*, 188 81-100. DOI: 10.1016/bs.pmbts.2021.10.002.

Crxxom. (2025, May 21). Daily Google News [Data set]. Kaggle. <https://www.kaggle.com/datasets/crxxom/daily-google-news>

Fridman, A., et al. (2021). COVID-19 and vaccine hesitancy: A longitudinal study. *PLOS ONE*, 16(4), DOI: 10.1371/journal.pone.0250123.

Gonen, H., et al. (2020). Simple, interpretable and stable method for detecting words with usage change across corpora. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* (pp. 538–555). DOI: 10.18653/v1/2020.acl-main.51.

Hamilton, W. et al. (2016). Diachronic word embeddings reveal statistical laws of semantic change. *arXiv*. 1605.09096

Kerr, J., et al. (2021). Political polarization on COVID-19 pandemic response in the United States. *Personality and Individual Differences*, 179, 110892. DOI: 10.1016/j.paid.2021.110892.

Krishnayah, J. (2023, July 17). *4.5M headlines from 2007–2022 (10 largest sites)* [Data set]. Kaggle. <https://www.kaggle.com/datasets/jordankrishnayah/45r-headlines-from-2007-2022-10-largest-sites>

Mikolov, T., et al. (2013). Efficient estimation of word representations in vector space. *arXiv*. 1301.3781

Monroe, B. L., et al. (2017). Fightin' Words: Lexical Feature Selection and Evaluation for Identifying the Content of Political Conflict. *Political Analysis*, 16(4), 372–403. DOI:10.1093/pan/mpn018

Motta, M., et al. (2020). How right-leaning

media coverage of COVID-19 facilitated the spread of misinformation in the early stages of the pandemic in the U.S. *Canadian Journal of Political Science*, 53(2), 335–342. DOI: 10.1017/S0008423920000396.

Pedregosa, F. et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.

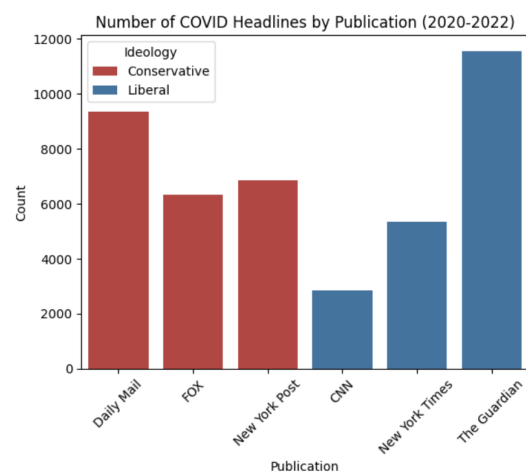
Sacerdote, B., et al. (2020). *Why is all COVID-19 news bad news?* (NBER Working Paper No. 28110). National Bureau of Economic Research. DOI: 10.3386/w28110.

Salton, G., & Buckley, C. (1988). Term-weighting approaches in automatic text retrieval. *Information Processing & Management*, 24(5), 513–523. DOI: 10.1016/0306-4573(88)90021-0.

Xu, M., et al. (2022). Media bias and factors affecting the impartiality of news agencies during COVID-19. *Behavioral Sciences*, 12(9), 313.

Appendices

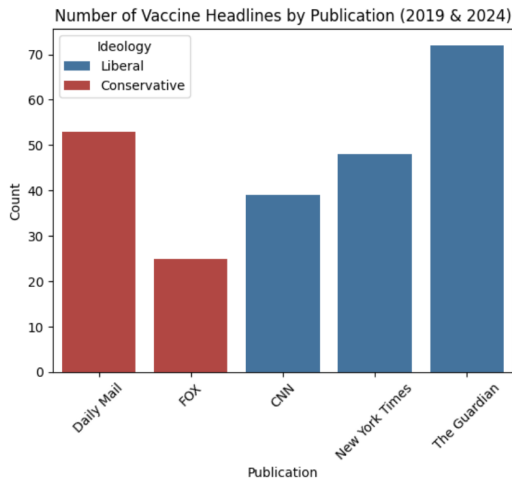
Appendix A. Number of COVID Headlines (2020-2022)



The dataset for addressing RQ1 and RQ2 contains 22,520 and 19,741 headlines from conservative and liberal news outlets, respectively. Of the conservative headlines, the Daily Mail published 9,351, the New York Post published 6,851, and

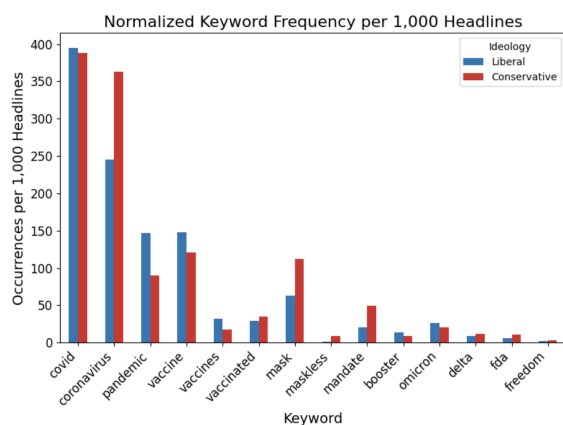
Fox published 6,318. Of the liberal headlines, The Guardian published 11,548, the New York Times published 5,345, and CNN published 2,848.

Appendix B. Number of Vaccine-related Headlines (2019 & 2024)



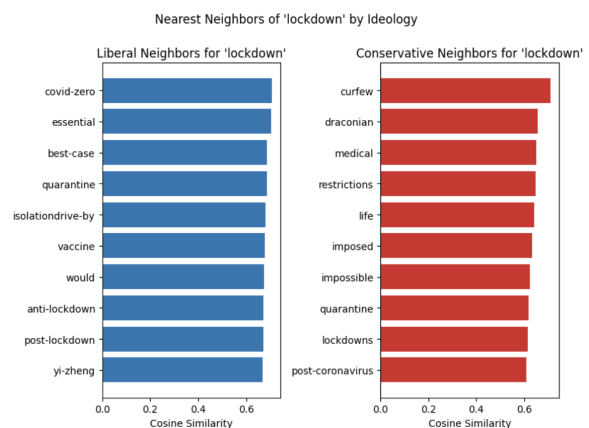
The dataset for addressing RQ3 contains 48 and 69 headlines from conservative and liberal news outlets in 2019, respectively. It also contains 30 and 90 headlines from conservative and liberal outlets in 2024, respectively. Of the conservative headlines from 2019 and 2024, the Daily Mail published 53 and Fox published 25. Of the liberal headlines from 2019 and 2024, CNN published 39, the New York Times published 48, and The Guardian published 72.

Appendix C. Normalized Keyword Frequency per 1,000 in the Pandemic Dataset (RQ1 & RQ2)



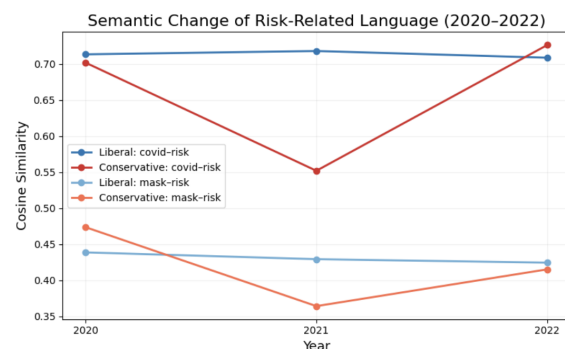
This figure displays the normalized frequency of select covid- and vaccine-related keywords in conservative vs. liberal headlines (per 1,000 headlines). Conservative outlets used words such as “coronavirus”, “mask”, and “mandate” at higher rates. Liberal outlets used words such as “pandemic”, “vaccine”, and “vaccines” at higher rates.

Appendix D. Nearest Neighbors of "lockdown"



Liberal outlets tend to cover lockdowns in logistical or health-focused contexts (e.g., "quarantine," "essential," "post-lockdown") whereas conservative outlets tend to associate lockdowns with governmental overreach and are more critical of them (e.g., "curfew," "draconian," "imposed").

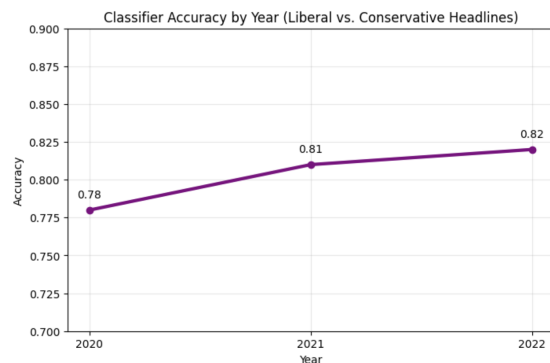
Appendix E. Semantic Change of Risk-Related Language



The cosine similarities for liberal outlets between covid-risk and mask-risk across all three years are relatively stable, reflecting this ideology's tendency to frame COVID in terms of public-health

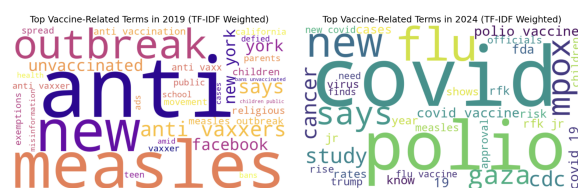
risk and masks as a way to decrease risk consistently throughout the pandemic. Conservative outlets show a sharp decline in 2021, likely reflecting a shift in coverage away from medical risk and toward political controversies and governmental overreach.

Appendix F. Ideological-Classifier Accuracy by Year



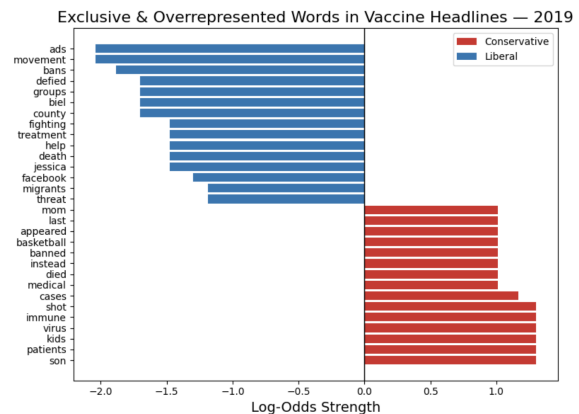
As mentioned in the paper, the classifiers' accuracies in distinguishing between liberal and conservative headlines increase from 0.78 to 0.81 to 0.82 across 2020 to 2022, suggesting that ideological differences in language became more pronounced as the pandemic progressed.

Appendix G. How Vaccine Language in Media Changed (2019 vs 2024)

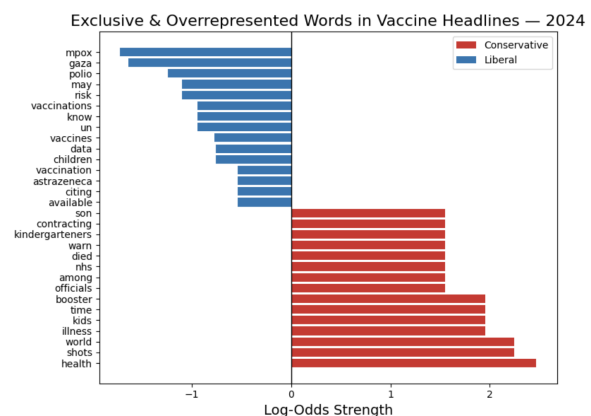


Pre-pandemic headlines focused on measles outbreaks and the anti-vaccination movement, reflecting concerns about misinformation. Post-pandemic headlines instead focused on COVID, flu, polio, and mpox. This suggests that the pandemic resulted in a shift in vaccine coverage away from misinformation and vaccine refusal to ongoing infectious disease threats.

Appendix H. Exclusive and Overrepresented Words in Vaccine Headlines (2019 vs 2024)

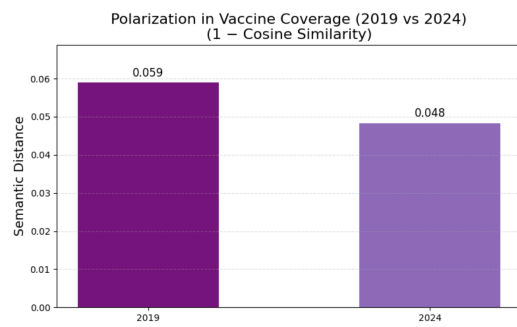


Pre-pandemic, liberal outlets concentrated on health measures and community-level impacts, while conservative outlets concentrated on emotional narratives.



Post-pandemic, liberal outlets emphasized global outbreaks and scientific-reporting, whereas conservative outlets emphasized skepticism on vaccines.

Appendix I. Polarization in Vaccine-related Headlines



There is slightly greater polarization in 2019 compared to 2024, suggesting that polarization regarding vaccines decreased after the pandemic.